

SalesGenie AI - Team B

API Contract, Tech Stack & Repository Setup — Day 2 Deliverables

1. Tech Stack Finalization

Layer	Technology
Backend / API	FastAPI (Python), Pydantic for schema validation
Database	PostgreSQL, SQLAlchemy ORM
AI Layer	Groq API (LLM inference)
Frontend	Streamlit, Plotly/Altair for charts
DevOps	Docker, GitHub Actions (CI/CD), monitoring via basic health-check endpoints
Testing	Pytest (backend), Postman (API contract verification)

2. Repository & Branching Setup

Repository Structure (as implemented on GitHub)

- SALESGENIE/ (repository root)
 - database/
connection.py (DB connection),
models.py (SQLAlchemy models)
 - modules/

module1_leads.py — Lead Management & Prospect Database

module2_intelligence.py — Lead Intelligence & Company Analysis

module3_outreach.py — AI Outreach Generation

module4_scoring.py — Lead Scoring & Recommendation Engine

module5_conversation.py — Conversation Intelligence & CRM Integration

module6_dashboard.py — Dashboard & Sales Analytics

- app.py — Streamlit frontend entry point
- main.py — FastAPI backend entry point (mounts routers from modules/)
- seed_data.py — sample data
- run_all.py — convenience script to run backend + frontend together
- requirements.txt
- README.md
- .env / .env.example

Branching Strategy

File / Area	Feature Branch	Primary Owner
database/ (connection.py, models.py)	feature/database-schema	Database / DevOps Engineer
modules/module1_leads.py	feature/module1-leads	Backend Developer
modules/module2_intelligence.py	feature/module2-intelligence	AI/ML Engineer
modules/module3_outreach.py	feature/module3-outreach	AI/ML Engineer
modules/module4_scoring.py	feature/module4-scoring	AI/ML Engineer
modules/module5_conversation.py	feature/module5-conversation	AI/ML Engineer
modules/module6_dashboard.py	feature/module6-dashboard	Frontend Developer + DevOps
app.py (Streamlit frontend)	feature/frontend-app	Frontend Developer
main.py (router mounting)	feature/backend-main	Backend Developer

- main — always deployable; protected, merges only via reviewed PR, only from develop at milestone checkpoints.
- develop — shared integration branch; all feature branches merge here first.
- Each feature branch is created from develop, and merged back into develop via PR once the module's task for the day/milestone is working.
- bugfix/<short-desc> — for fixing a bug found during testing (e.g. bugfix/module4-score-rounding).
- hotfix/<short-desc> — for urgent fixes needed directly on main after a milestone deploy.
- QA/Test Engineer tests each feature branch (or develop after merge) rather than owning a dedicated branch.

3. API Contract — Endpoint List

module1_leads.py — Lead Management & Prospect Database

Method / Endpoint	Description	Request Body (key fields)
POST /leads	Create a new lead	company_name, industry, contact_name, email, phone
GET /leads	List / search prospects (filters, pagination)	query params: q, industry, status, page
GET /leads/{lead_id}	Get lead detail	—
PUT /leads/{lead_id}	Update a lead record	any editable lead fields
DELETE /leads/{lead_id}	Delete a lead record	—

module2_intelligence.py — Lead Intelligence & Company Analysis

Method / Endpoint	Description	Request Body (key fields)
POST /leads/{lead_id}/analyze	Trigger AI company-profile analysis	— (reads stored company data)
GET /leads/{lead_id}/insights	Get generated company insights	—
POST /leads/{lead_id}/qualify	Generate qualification / lead score	—
GET /leads/{lead_id}/score	Get qualification score & factors	—

module3_outreach.py — AI Outreach Generation

Method / Endpoint	Description	Request Body (key fields)
POST /outreach/generate	Generate a personalized outreach email	lead_id, tone/industry template
GET /outreach/{campaign_id}	Get a generated campaign draft	—
PUT /outreach/{campaign_id}	Edit a draft before sending	email_subject, email_content
POST /outreach/{campaign_id}/send	Mark / dispatch campaign as sent	—

module4_scoring.py — Lead Scoring & Recommendation Engine

Method / Endpoint	Description	Request Body (key fields)
POST /leads/{lead_id}/conversion-score	Predict conversion probability	—
GET /leads/{lead_id}/recommendations	Get next-best-action recommendations	—

module5_conversation.py — Conversation Intelligence & CRM Integration

Method / Endpoint	Description	Request Body (key fields)
POST /crm/sync	Trigger CRM sync (contacts, activities, deals)	crm_platform, lead_id(s)
GET /crm/status	Get CRM sync status / logs	—
POST /conversations/summarize	Summarize a call/meeting transcript	lead_id, transcript_text or audio_ref
GET /conversations/{lead_id}	Get summaries & action items for a lead	—

module6_dashboard.py — Dashboard & Sales Analytics

Method / Endpoint	Description	Request Body (key fields)
GET /analytics/dashboard	Aggregate KPIs (conversion rate, pipeline value, response time)	query params: period
GET /analytics/followups	Automated follow-up recommendation feed	—